

# Partial result for arXiv:1301.5895

## The covering-pessimum conjecture for two ranges of superballs

**Status. partial result.** We prove the source's global conjecture throughout two infinite subfamilies of three-dimensional superballs. The result is not claimed for arbitrary symmetric bodies, and a middle exponent interval remains.

**Source conjecture.** Kallus proves that the Euclidean ball is a strict local maximum of lattice covering density among origin-symmetric bodies and ends with the following global conjecture:

where  $\Psi$  is the convolution operator in Lemma 4. Since  $\int \Psi[\rho] a \sigma = 0$  and  $\Phi[|\rho|]$  is non-negative, we have that  $\min(-\frac{1}{8}\Phi[\rho] + \varepsilon'\Phi[|\rho|]) \leq -\frac{1}{16}\|\Phi[\rho]\|_1 + \varepsilon'\|\Phi[|\rho|]\|_1$ , and so

$$\Delta_K \leq -\frac{1}{16}\|\Phi^{-1}\|^{-1} \cdot \|\rho\|_1 + (\varepsilon'\|\Phi\| + \varepsilon'')\|\rho\|_1.$$

Therefore, we conclude that there is a coefficient  $c > 0$  such that  $\Delta_K < -c\|\rho\|_1$ .  $\square$

Theorem 4 and the fact that the disk is pessimal for covering in the plane motivate the following conjecture:

**Conjecture.** For all origin-symmetric convex bodies  $K \subseteq \mathbb{R}^3$  that are not ellipsoids,  $\vartheta(K) < \vartheta(B^3)$ .

For  $1 \leq p < \infty$ , put

$$B_p^3 = \left\{ x \in \mathbb{R}^3 : \sum_{i=1}^3 |x_i|^p \leq 1 \right\}, \quad v_p = \text{vol}(B_p^3) = \frac{8\Gamma(1+1/p)^3}{\Gamma(1+3/p)}.$$

The Euclidean ball has optimal lattice covering density

$$\vartheta(B_2^3) = \frac{5\sqrt{5}\pi}{24} = 1.463503\dots$$

**Theorem.** For every  $1 \leq p < 2$  and every  $p \geq 9$ ,

$$\vartheta(B_p^3) < \vartheta(B_2^3).$$

At  $p = 2$  equality holds, as all ellipsoids are affine images of the ball.

## The BCC certificate for $1 \leq p \leq 2$

Let

$$\Lambda_{\text{BCC}} = \mathbb{Z}^3 \cup (\mathbb{Z}^3 + h), \quad h = (1/2, 1/2, 1/2).$$

It is a lattice of determinant  $1/2$ .

**Lemma.** For every  $p \geq 1$ , the  $\ell_p$  covering radius of  $\Lambda_{\text{BCC}}$  is exactly

$$r_p = \frac{1}{2}(1 + 2^{-p})^{1/p}.$$

*Proof.* Reduce a point modulo  $\mathbb{Z}^3$  and reflect its coordinates. Reflections preserve both cosets, so the point is represented by  $z/2$  with  $z \in [0, 1]^3$ . Its distance to the lattice is at most

$$\frac{1}{2} \min \left\{ \left( \sum z_i^p \right)^{1/p}, \left( \sum (1 - z_i)^p \right)^{1/p} \right\}.$$

At least one of  $\sum z_i$  and  $\sum (1 - z_i)$  is at most  $3/2$ . The convex function  $\sum z_i^p$  attains its maximum on  $\{z \in [0, 1]^3 : \sum z_i \leq 3/2\}$  at an extreme point; the largest value is  $1 + 2^{-p}$ , attained at permutations of  $(1, 1/2, 0)$ . This proves the upper bound. At  $z = (1, 1/2, 0)$ , coordinatewise nearest-point inspection in both cosets shows that the displayed two distances are equal to  $r_p$  and no other lattice point is closer. Hence the bound is exact.  $\square$

The corresponding covering density is

$$D_{\text{BCC}}(p) = \frac{v_p r_p^3}{\det \Lambda_{\text{BCC}}} = \frac{v_p}{4} (1 + 2^{-p})^{3/p}. \quad (1)$$

**Lemma.**  $D_{\text{BCC}}(p)$  is strictly increasing on  $1 \leq p \leq 2$ .

*Proof.* Set  $t = 1/p$ ,  $a = 2^{-1/t}$ , and differentiate the logarithm of (1). With  $\psi = \Gamma'/\Gamma$  one obtains

$$\frac{1}{3} \frac{d}{dt} \log D_{\text{BCC}}(1/t) = \psi(1+t) - \psi(1+3t) + R(t), \quad R(t) = \log(1+a) + \frac{a \log 2}{t(1+a)}.$$

The series for the trigamma function and comparison with its integral give

$$\psi(1+3t) - \psi(1+t) = \int_{1+t}^{1+3t} \psi'(u) du > \log \frac{1+3t}{1+t}.$$

It remains to check  $H(t) := \log \frac{1+3t}{1+t} - R(t) > 0$  on  $[1/2, 1]$ . Differentiation gives

$$H'(t) = \frac{2}{(1+t)(1+3t)} - \frac{(\log 2)^2 a}{t^3(1+a)^2},$$

whose sign is the sign of

$$Q(t) = 2t^3(1+a)^2 - (\log 2)^2 a(1+t)(1+3t).$$

Direct differentiation shows  $Q'(t) > 0$  on  $[1/2, 1]$ . Moreover  $Q(.55) < 0 < Q(.56)$ ,  $H(.55) > .0082$ ,  $H(.56) > .0081$ , and on  $[.55, .56]$ , monotonicity of the two terms gives

$$H(t) \geq \log \frac{1+3(.55)}{1+.55} - R(.56) > .0033.$$

Thus  $H > 0$ . The decimal inequalities are certified by directed interval arithmetic in the included verifier (which also certifies  $Q' > 0$  on 5,000 subintervals); they merely abbreviate an elementary one-variable check. Therefore the displayed logarithmic derivative is negative, which is equivalent to strict increase in  $p$ .  $\square$

At  $p = 2$ , formula (1) becomes exactly

$$\frac{4\pi/3}{4} \left(1 + \frac{1}{4}\right)^{3/2} = \frac{5\sqrt{5}\pi}{24}.$$

The monotonicity lemma proves the theorem for  $1 \leq p < 2$ .

## The cubic certificate for $p \geq 9$

The covering radius of  $\mathbb{Z}^3$  in  $\ell_p^3$  is  $3^{1/p}/2$ , attained at the center of its unit cube. Hence

$$\vartheta(B_p^3) \leq \frac{v_p 3^{3/p}}{8} \leq 3^{3/p},$$

because  $B_p^3 \subset [-1, 1]^3$  and therefore  $v_p \leq 8$ . For  $p \geq 9$  this is at most  $3^{1/3} < 1.443$ , whereas

$$\frac{5\sqrt{5}\pi}{24} > \frac{5(2.236)(3.1415)}{24} > 1.4634.$$

This proves the second range.

## Upgrade audit and limitations

Eight focused attacks are detailed in the accompanying attempt note. In particular, exact boundary checks show why the BCC lattice ceases to certify the result for  $p > 2$ : the witness  $(1, 1/2, 0)$  persists and its density rises above the ball value. A rhombohedral numerical search initially looked better, but auditing its faces exposed omitted covering holes. Cubic and planar-product lattices also leave a gap.

Thus this packet does not settle  $2 < p < 9$ , arbitrary origin-symmetric bodies, or the full global conjecture. The source's local theorem does cover some unspecified neighborhood of  $p = 2$ ; the new contribution here is a global certificate along the whole lower superball branch and an elementary far-end certificate on the upper branch.

**Verification.** The exact cell argument is independent of computation. The included script provides directed-interval certification of the only scalar calculus lemma, checks half a million cell points and all equality witnesses used by the formula, and verifies density sanity values. A bounded title/exact-conjecture search found no later full resolution.

## References

- [1] Y. Kallus, *When is the ball a local pessimum for covering?*, Discrete Comput. Geom. **54** (2015), 232–245; arXiv:1301.5895.